NEURAL NETWORK : Model the brain.

① Why model.

$$\boxed{a_i} \longrightarrow \boxed{X} \longrightarrow \boxed{a_j}$$

$$\begin{array}{cc} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \\ 0 & 0 \end{bmatrix} \\ a_{i_1} \quad a_{i_2} \end{array} \longrightarrow \boxed{X} \longrightarrow \begin{array}{c} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \\ a_j \end{array}$$

- To see how X behaves.
  - Cannot be determined analytical.
  - You have a specific hypothesis H_0 about X or how systems like X (e.g. brain, people etc.) behave.

How?

- Intro, motivation $\rightarrow H_0$
- Model design
  - architecture
  - activation function
  - learning function
- Simulations
- Discuss, conclusion.

PERCEPTRON architecture.

Notations

i^{th} layer: input layer

j^{th} layer: output layer

k^{th} unit

non-linear activation function.

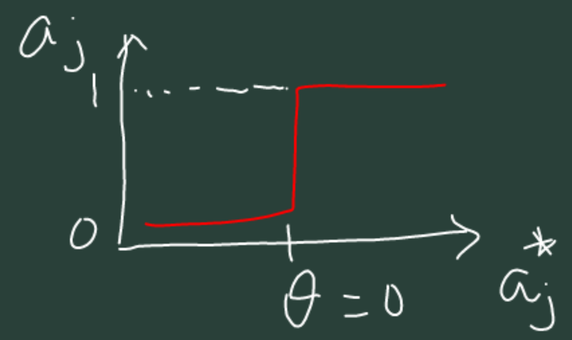
$a_j = a_0$

Activation Function

$$a_j^* = \sum_k a_{i_k} w_{i_k j}$$

- Preactivation : $\boxed{\vec{a}_i W = a_j^*}$ * — (1)

- activation, non-linear : $a_j = \max(\text{sign}(a_j^*), \underset{\theta}{\underset{\shortparallel}{0}})$



or, $\boxed{a_j = \begin{cases} 1 & \text{if } a_j^* > \theta \\ 0 & \text{otherwise} \end{cases}}$ * — (2)

$$a_j = \max(\text{sign}(a_j^* - \theta), 0) \quad \text{— (3)}$$

$$\underbrace{a_j = w_{i_1 j} \cdot a_{i_1} + w_{i_2 j} \cdot a_{i_2}}_{\text{General } \underline{\text{Linear}} \text{ Model}}$$

$\underline{\text{linear}}$ regression

What if $a_j = \vec{a}_i W$

rather than

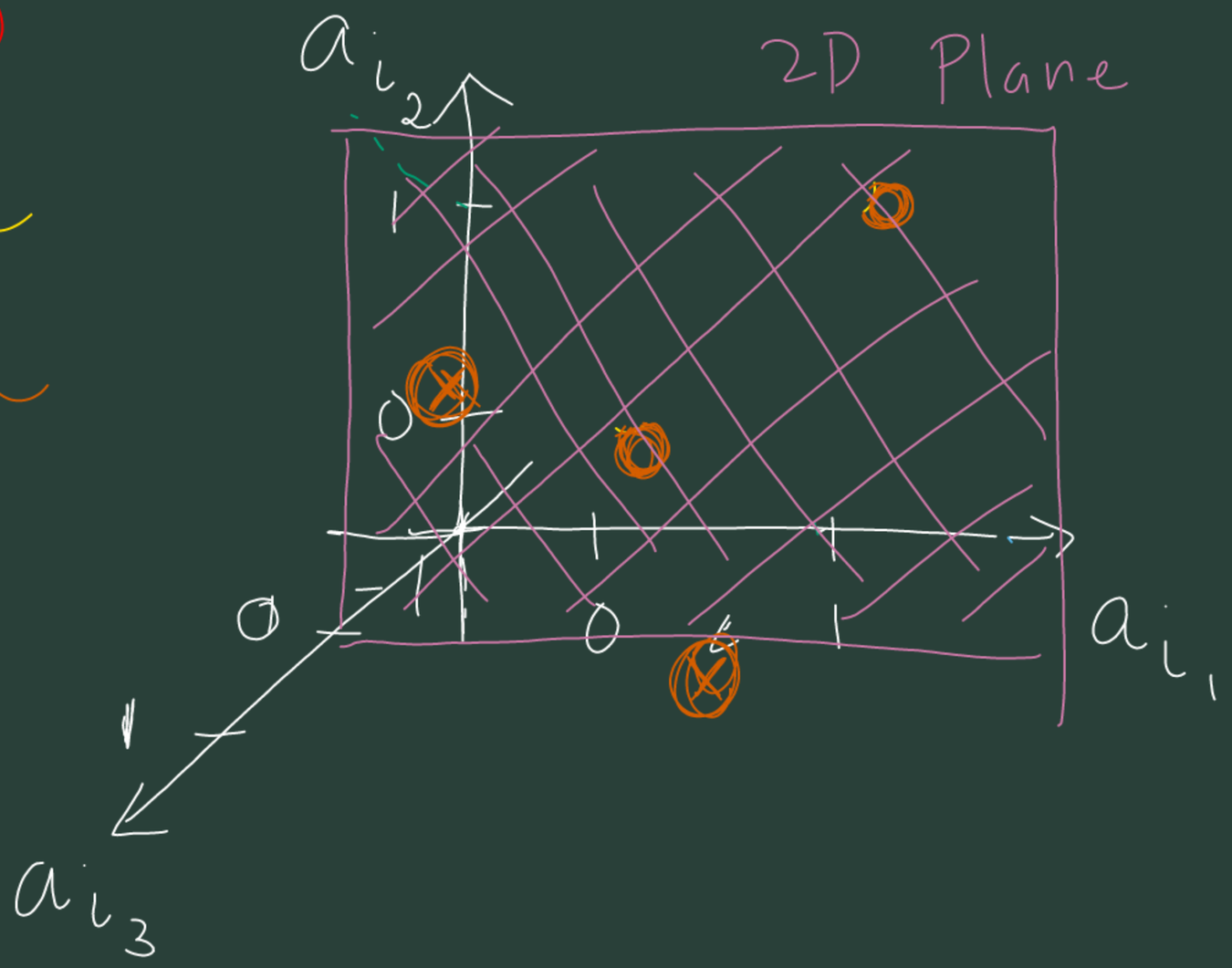
$$a_j = \begin{cases} 1 & \text{if } \vec{a}_i W > \theta \\ 0 & \text{otherwise} \end{cases}$$

applies non-linear component.

AND

OR

XOR



$a_j = 1$, ⊗

$a_j = 0$, ○